



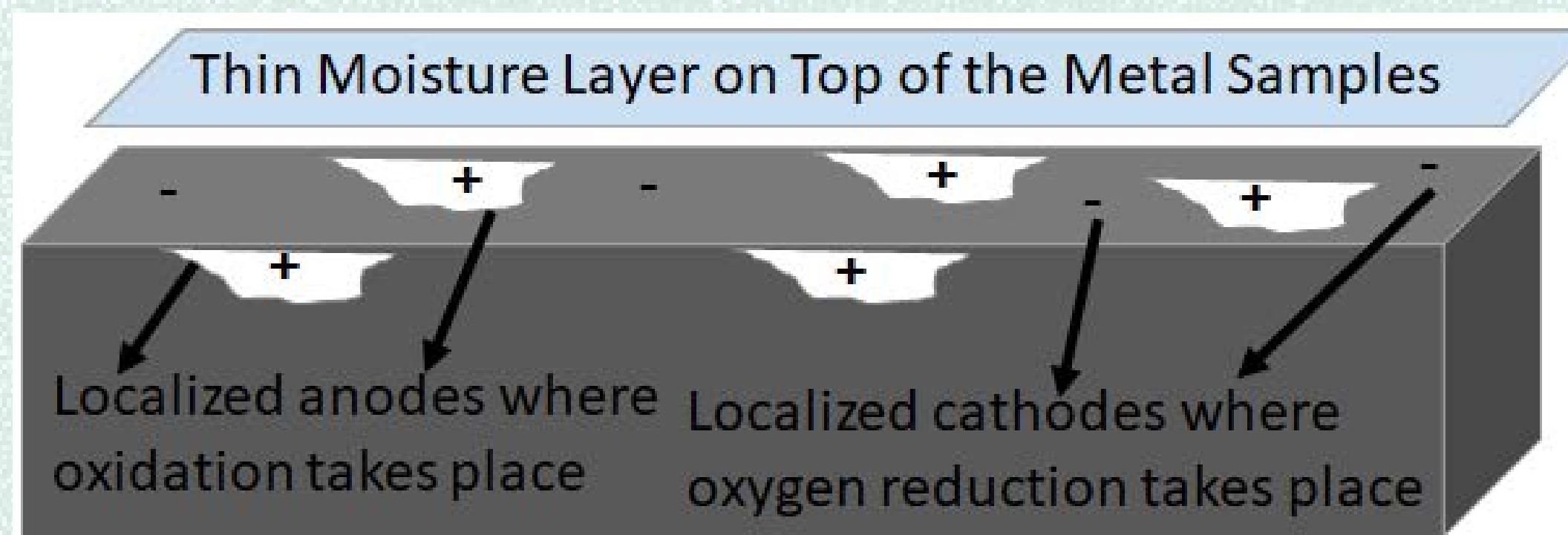
Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska

Raghu Srinivasan, Lawrence Giron Jr and Jozef Hunter, College of Engineering, University of Alaska Anchorage

INTRODUCTION

- Atmospheric corrosion is an important form of corrosion because many structures and systems are used in natural environments.
- Atmospheric corrosion is a complex process, which involves chemical, electrochemical, and physical changes to the metal exposed.

For atmospheric corrosion, the electrolyte is a thin layer of moisture (not completely immersed in solution) and the corrosion occurs at the metal/electrolyte interface when exposed to meteorological or environmental factors.



OBJECTIVES

- To assemble and deploy multi-angle corrosion test racks with auxiliary weather station at four test sites across Alaska,
- To monitor and measure the ambient air temperature, relative humidity, time of wetness at exposure angles of 0°, 30°, and 45°, total rainfall, and aerosol chloride and sulfate deposition at the test site across 6-month, 12-month, and 24-month exposure periods.
- To expose five different metal alloys to atmospheric conditions in Alaska at 0°, 30°, and 45° exposure angles.
- To characterize Alaska's environmental severity and categorize corrosivity with determined corrosion rates through exposure of specimens and the estimation of corrosion rates through dose-response functions and weather parameters

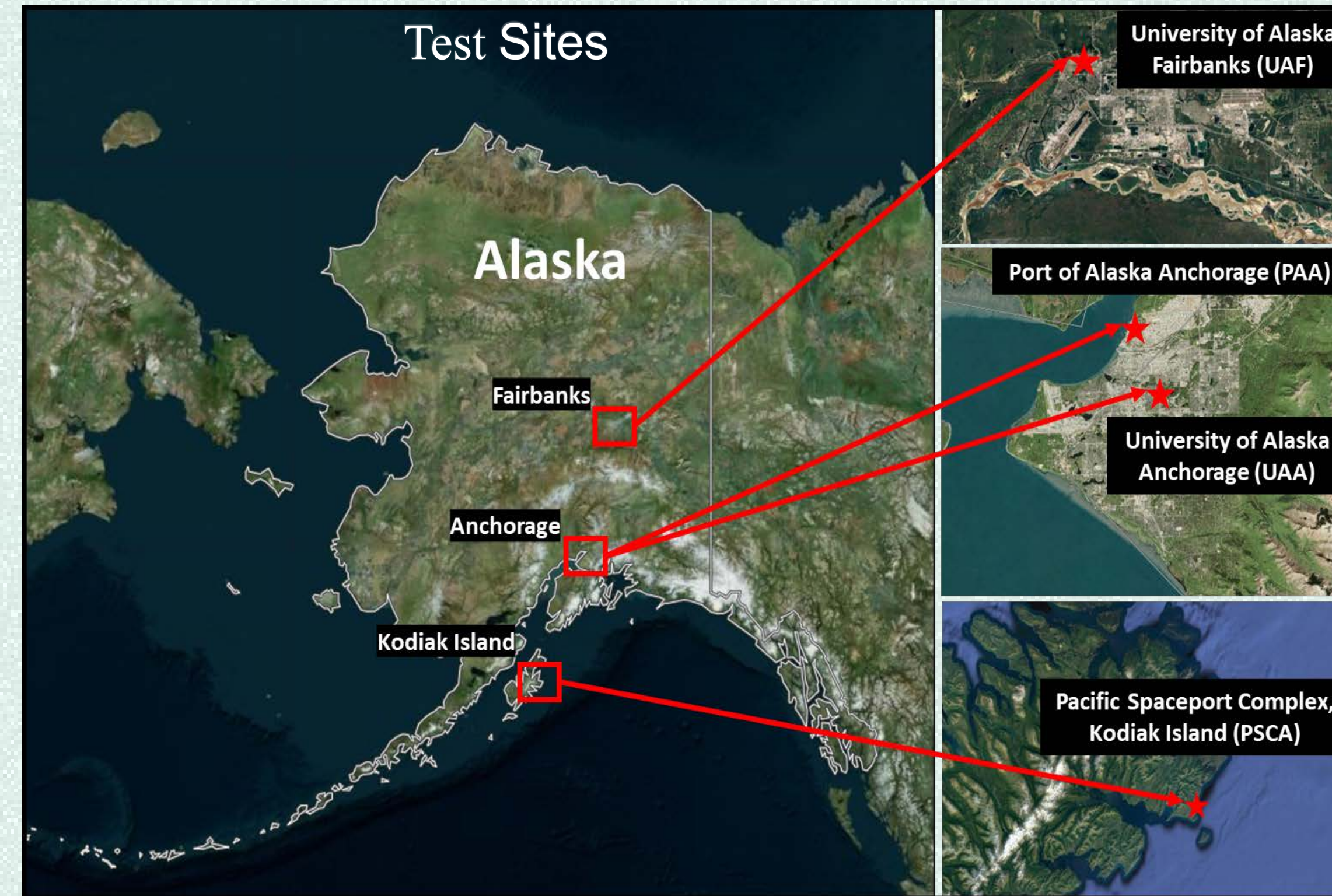


Table 1: Test Site Coordinates, Distance from Salt Water, Elevation, and Environment

Location	Coordinates	Distance to Nearest Body of Salt Water	Elevation AMSL	Environment
Pacific Spaceport Complex – Alaska	57.453°N, 152.346°W	~600 feet (183 meters) from Pacific Ocean	0 feet (0 meters)	Aggressive Marine Environment
University of Alaska Fairbanks	64.855°N, 147.817°W	~250 miles (400 km) from Pacific Ocean ~350 miles (560 km) from Arctic Ocean	60–72 feet (18–22 meters)	Far-inland Urban Environment
Port of Alaska	61.239°N, 149.888°W	~30 feet (9 meters) from Pacific Ocean	30–36 feet (18–22 meters)	Moderate Marine Environment
University of Alaska Anchorage	61.192°N, 149.822°W	~4 miles (6.4 km) from Pacific Ocean	30–45 feet (9–13 meters)	Inland Mild-Marine Environment

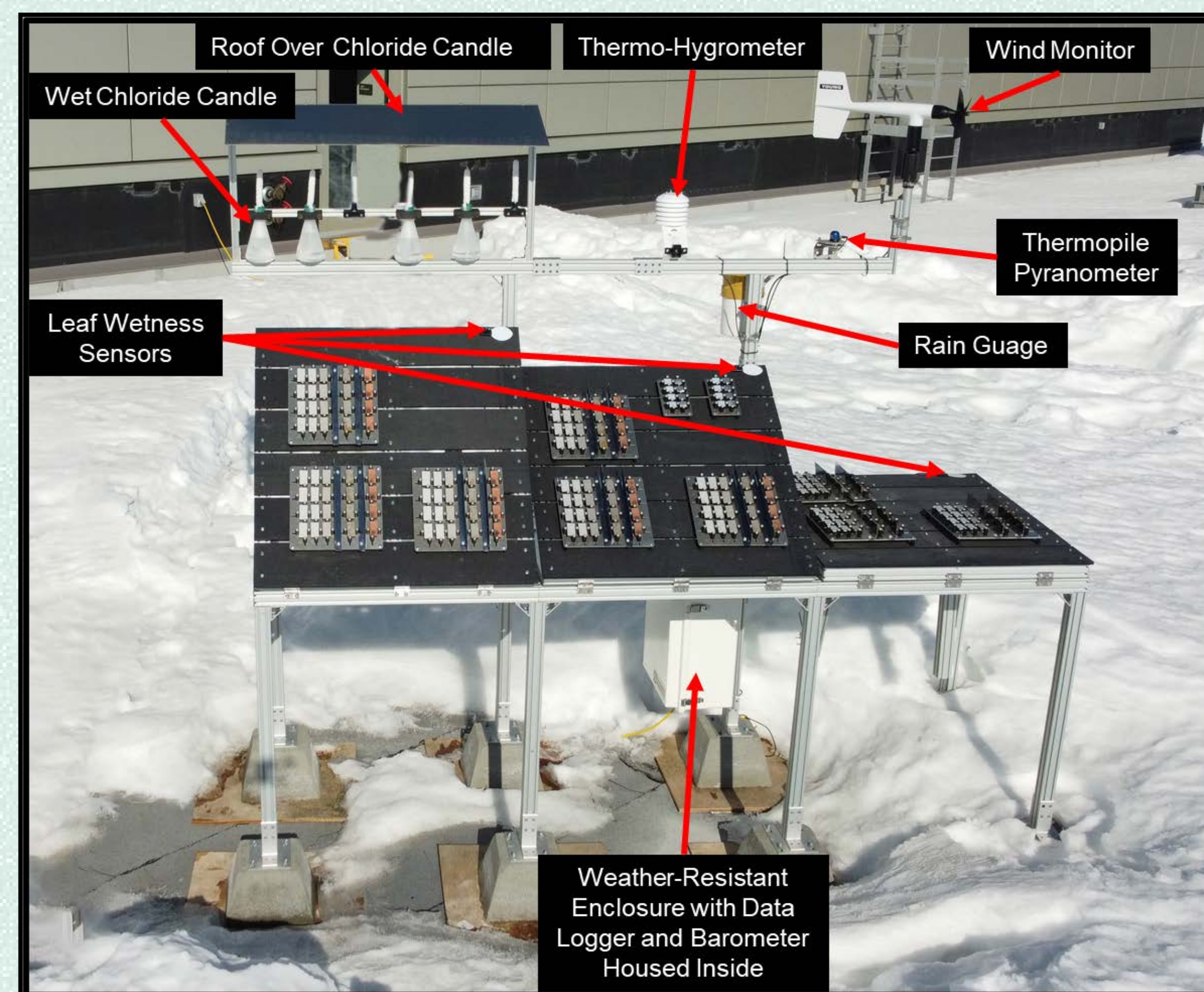
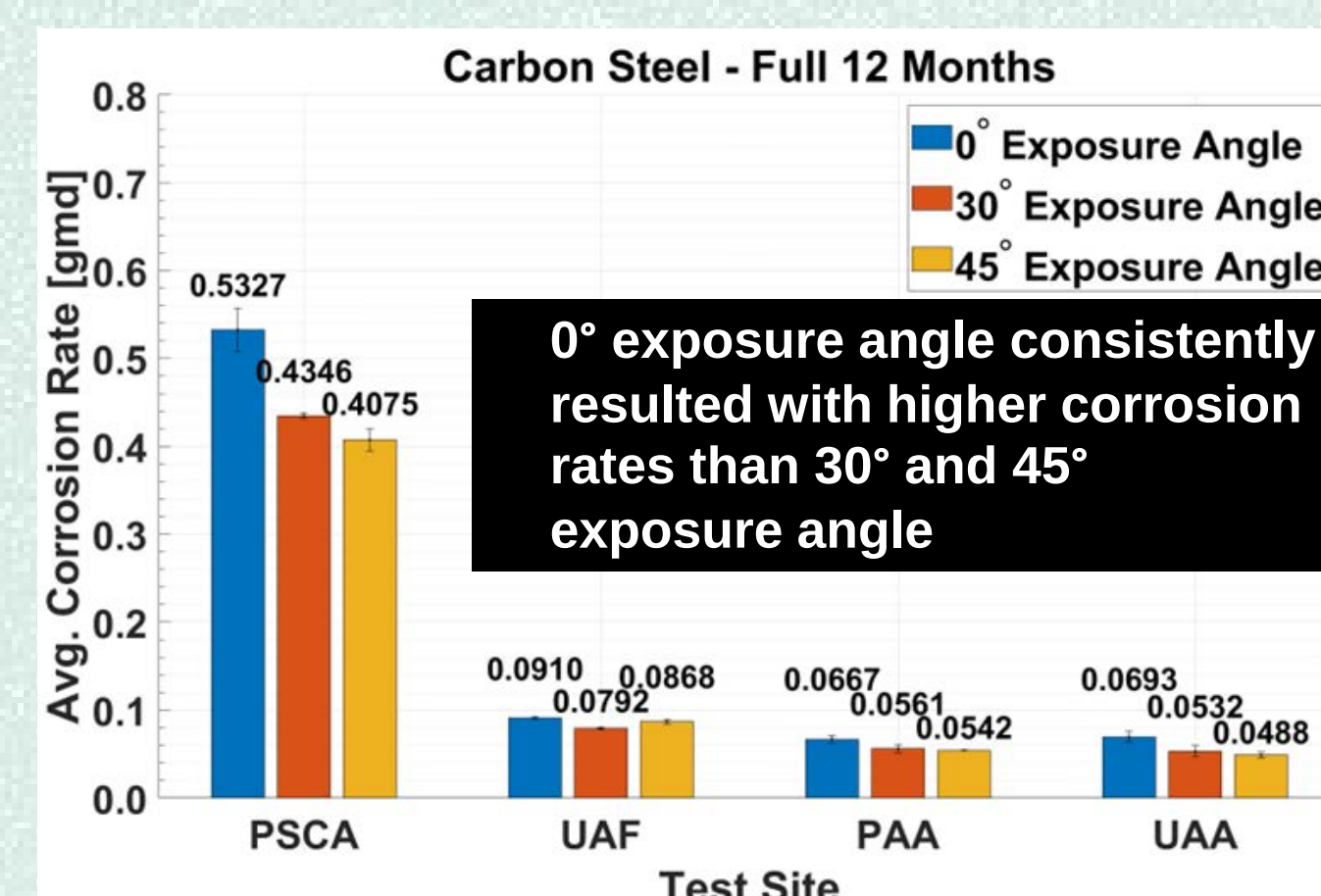


Table 3: Summary of Weather Data at each Test Site over the First 12-Month Exposure Period

Location	Average Air Temperature [°C]	Average Relative Humidity [%]	Time of Wetness [%]			Chloride Deposition [mg/(m²×day)]	Sulfate Deposition [mg/(m²×day)]
			0°	30°	45°		
PSCA	6.26	85	54.2	48.2	41.4	95.4	20.3
UAF	-0.07	66.8	17.7	13.8	10.9	18.5	6
Port of Alaska	8.86	77.1	30	25.4	23.4	22.7	5
UAA	9.64	76.3	—	—	—	17.7	3.8



- Carbon steel coupons exposed at the PSCA test site exhibited corrosion rates four times greater than the carbon steel coupons exposed at the UAF, PAA, and UAA test sites.
- This contrast can be attributed to the weather where the PSCA test site recorded four times the chloride deposition rate and 18% more TOW.

CORROSSIVITY CATEGORIZATION

Category	Corrosivity
C1	Very low
C2	Low
C3	Medium
C4	High
C5	Very high
CX	Extreme

- Each test location and angle is classified within the C2 (low) corrosivity category

Corrosivity category	Corrosion rates of metals			
	Unit	Carbon steel	Zinc	Copper
C1	g/(m²·a) µm/a	$r_{corr} \leq 10$ $r_{corr} \leq 1.3$	$r_{corr} \leq 0.7$ $r_{corr} \leq 0.1$	$r_{corr} \leq 0.9$ $r_{corr} \leq 0.1$
C2	g/(m²·a) µm/a	$10 < r_{corr} \leq 200$ $1.3 < r_{corr} \leq 25$	$0.7 < r_{corr} \leq 5$ $0.1 < r_{corr} \leq 0.7$	$0.9 < r_{corr} \leq 5$ $0.1 < r_{corr} \leq 0.6$
C3	g/(m²·a) µm/a	$200 < r_{corr} \leq 400$ $25 < r_{corr} \leq 50$	$5 < r_{corr} \leq 15$ $0.7 < r_{corr} \leq 2.1$	$5 < r_{corr} \leq 12$ $0.6 < r_{corr} \leq 1.3$
C4	g/(m²·a) µm/a	$400 < r_{corr} \leq 650$ $50 < r_{corr} \leq 80$	$15 < r_{corr} \leq 30$ $2.1 < r_{corr} \leq 4.2$	$12 < r_{corr} \leq 25$ $1.3 < r_{corr} \leq 2.8$
C5	g/(m²·a) µm/a	$650 < r_{corr} \leq 1500$ $80 < r_{corr} \leq 200$	$30 < r_{corr} \leq 60$ $4.2 < r_{corr} \leq 8.4$	$25 < r_{corr} \leq 50$ $2.8 < r_{corr} \leq 5.6$
CX	g/(m²·a) µm/a	$1500 < r_{corr} \leq 5000$ $200 < r_{corr} \leq 700$	$60 < r_{corr} \leq 180$ $8.4 < r_{corr} \leq 25$	$50 < r_{corr} \leq 90$ $5.6 < r_{corr} \leq 10$

Test Site	Metal: UNS G10080 Average First Year Corrosion Rate with Corresponding Category		
	Angle	r_{corr} [g/m²/year]	Corrosivity category
PSCA	0°→	194.4	~C3
	30°→	158.6	
	45°→	148.7	
UAF	0°→	33.2	C2
	30°→	28.9	
	45°→	31.7	
PAA	0°→	24.3	C2
	30°→	20.5	
	45°→	19.8	
UAA	0°→	25.3	C2
	30°→	19.4	
	45°→	17.8	

- PSCA site approaches the upper boundary of C2, with the 0° orientation verging on the C3 (medium) classification.

Corrosivity Categorization For 1008 Carbon Steel

$$r_{corr} = 1.77 \cdot P_d^{0.52} \cdot \exp(0.020 \cdot RH + f_{St}) + 0.102 \cdot S_d^{0.62} \cdot \exp(0.033 \cdot RH + 0.040 \cdot T)$$

$$f_{St} = 0.150 \cdot (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.054 \cdot (T - 10)$$

Symbol	Description	Interval	Unit
T	Temperature	-17,1 to 28,7	°C
RH	Relative humidity	34 to 93	%
P_d	SO ₂ deposition	0,7 to 150,4	mg/(m²·d)
S_d	Cl⁻ deposition	0,4 to 760,5	mg/(m²·d)

Test Site	Metal: UNS G10080 Comparison between determined and estimated 1 st year corrosion rates			
	Angle	r_{corr} [g/m²/year] (Determined through exposure of specimens)	Corrosivity Category	r_{corr} [g/m²/year] (Estimation through dose-response function)
PSCA	30°→	158.6	C2	494.6
UAF	30°→	28.9		73.8
PAA	30°→	20.5		146.0
UAA	30°→	19.4		119.9

The function does not take all comprehensive environmental factors into account including TOW, UV radiation, rainfall, wind velocity, and nitrogen oxides

Corrosivity Categorization For Aluminum and Copper Alloys

Test Site	Metal: UNS A92024 Comparison between determined and estimated 1 st year corrosion rates			
	Angle	r_{corr} [g/m²/year] (Determined through exposure of specimens)	Corrosivity Category	r_{corr} [g/m²/year] (Estimation through dose-response function)
PSCA	30°→	5.5115	C5	1.6113
UAF	30°→	1.0220	C3	0.3184
PAA	30°→	1.2045		0.4719
UAA	30°→	1.2045		0.3829

Test Site	Metal: UNS A96061 Comparison between determined and estimated 1 st year corrosion rates			
	Angle	r_{corr} [g/m²/year] (Determined through exposure of specimens)	Corrosivity Category	r_{corr} [g/m²/year] (Estimation through dose-response function)
PSCA	30°→	1.1680	C3	1.5650
UAF	30°→	0.8395		0.3093
PAA	30°→	0.803		0.4583
UAA	30°→	0.8395		0.3719

Test Site	Metal: UNS C11000 Comparison between determined and estimated 1 st year corrosion rates			
	Angle	r_{corr} [g/m²/year] (Determined through exposure of specimens)	Corrosivity Category	r_{corr} [g/m²/year] (Estimation through dose-response function)
PSCA	30°→	14.4905	C4	18.8613
UAF	30°→	5.2560	C3	3.3203
PAA	30°→	4.7815	C2	7.4647
UAA	30°→	4.4165		6.5520

$$r_{corr} = 0.0042 \cdot P_d^{0.73} \cdot \exp(0.025 \cdot RH + f_{Al}) + 0.0018 \cdot S_d^{0.60} \cdot \exp(0.020 \cdot RH + 0.094 \cdot T)$$

$$f_{Al} = 0.009 \cdot (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.043 \cdot (T - 10)$$

- Level of uncertainty through method of determination (exposure of specimens) is ±5% and through method of estimation (dose-response function) is -50% to +100% for aluminums
- Determined category of 2024-T3 AA through the exposure of specimens at PSCA was C5, but the estimated category was only a C3, medium corrosivity

$$r_{corr} = 0.0053 \cdot P_d^{0.26} \cdot \exp(0.059 \cdot RH + f_{Cu}) + 0.01025 \cdot S_d^{0.27} \cdot \exp(0.036 \cdot RH + 0.049 \cdot T)$$

$$f_{Cu} = 0.126 \cdot (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.080 \cdot (T - 10)$$

- For copper, the level of uncertainty through method of determination (exposure of specimens) is ±2% and through method of estimation (dose-response function) is -33% to +50%

- Newer standardized models are required to better estimate and predict corrosion in colder climates like Alaska.
- Create a dose-response function that represents the atmosphere of Alaska that incorporates the time of wetness and exposure angle to estimate metal alloys' corrosion rates in colder regions.